



LECTURE 9

Agile Principles - Revisited

COEN / ELEC 390

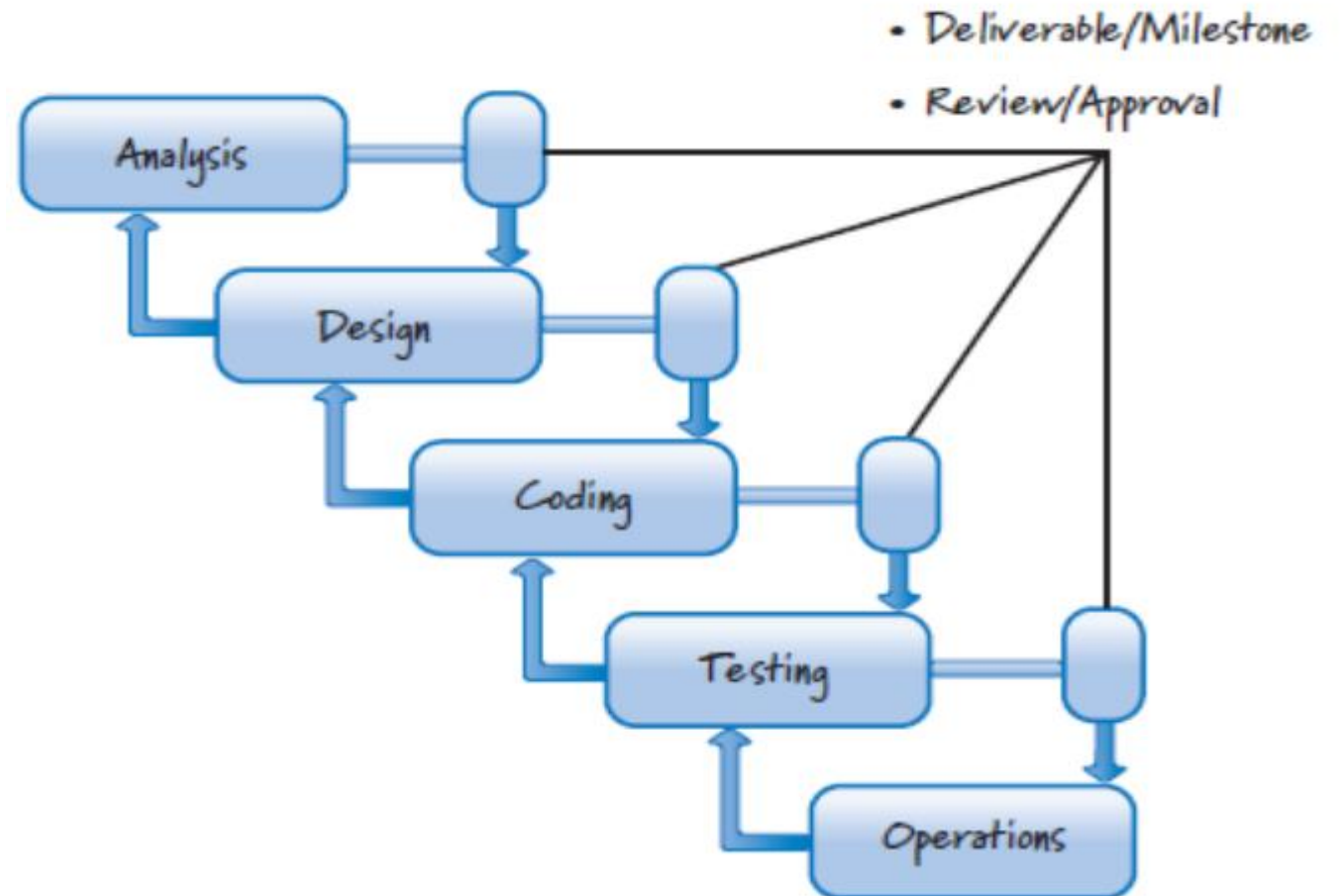
Acknowledgement

- The concepts and figures in these slides are based on the textbook:

Essential Scrum, A Practical Guide to the Most Popular Agile Process
Kenneth S. Rubin, Addison-Wesley, 2013.

Plan Driven Development Techniques – Waterfall Process

- Works well if the problem is
 - Well defined
 - Well known or Predictable
 - Not changing



Agile Principles

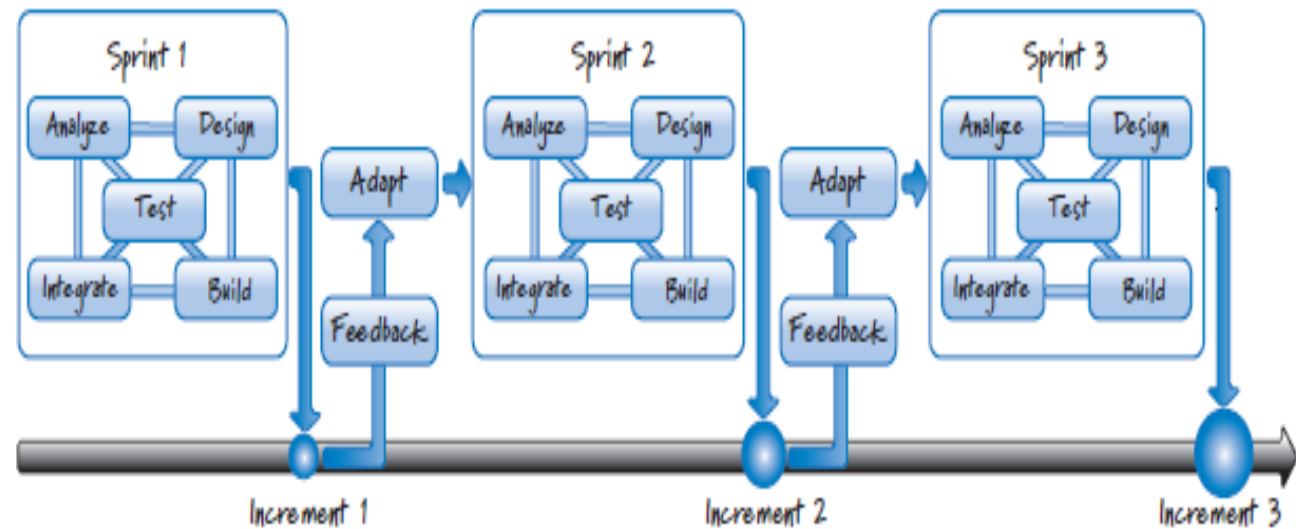
- Embrace helpful variability
- Employ iterative and incremental development.
- Leverage variability through inspection, adaptation, and transparency.
- Reduce all forms of uncertainty simultaneously.

Embrace Helpful Variability

- Recognize and accept that some variability in the process, team dynamics, and project requirements
- Instead of striving for a rigid approach, teams are encouraged to adapt and adjust their practices based on the unique context of their work
- Let's discuss some examples related to your project

Incremental and Iterative Process

- Each sprint should be a complete product development cycle with a potentially “shippable” product
- Thus we pass through and over the product many times, learning more, polishing more, perfecting more
- Don't work on a phase, but a feature; a feature should be integrated and tested.



Leverage Variability via **Inspection, Adaptation, and Transparency**

- Inspection
 - Consciously examine product and process
- Adaptation
 - Act and change based on inspection and other data
- Transparency
 - All information needed to produce the product must be available to those involved in creating it
 - Makes Inspection and adaptation possible.

Plan Driven vs. Scrum Processes

Dimension	Plan-Driven	Scrum
Degree of process definition	Well-defined set of sequential steps	Complex process that would defy a complete up-front definition
Randomness of output	Little or no output variability	Expect variability because we are not trying to build the same thing over and over
Amount of feedback used	Little and late	Frequent and early

Forms of Uncertainty

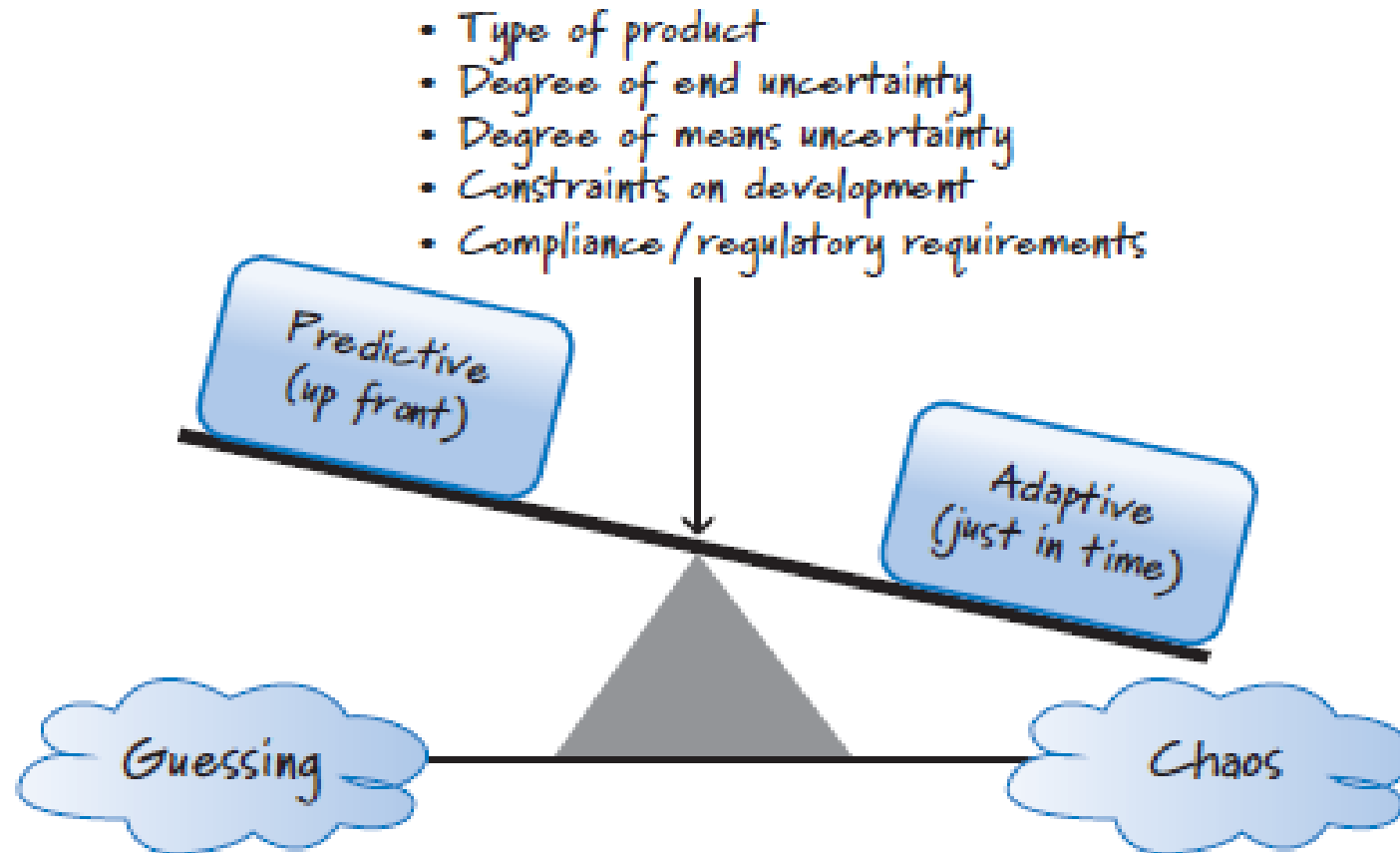
- **End Uncertainty** (*what* uncertainty): Uncertainty surrounding the features of the final product.
- **Means Uncertainty** (*how* uncertainty): uncertainty surrounding the process and technologies used to develop a product
- We should aim to reduce all forms of uncertainty as we progress through the project

Prediction and Adaptation

- Keep options open
- Accept that you cannot get it right up front
- Favor an adaptive, exploratory approach
- Embrace change in an economically sensible way
- Balance predictive up-front work with adaptive just-in-time work

Let's discuss how this applies to the design steps of your project

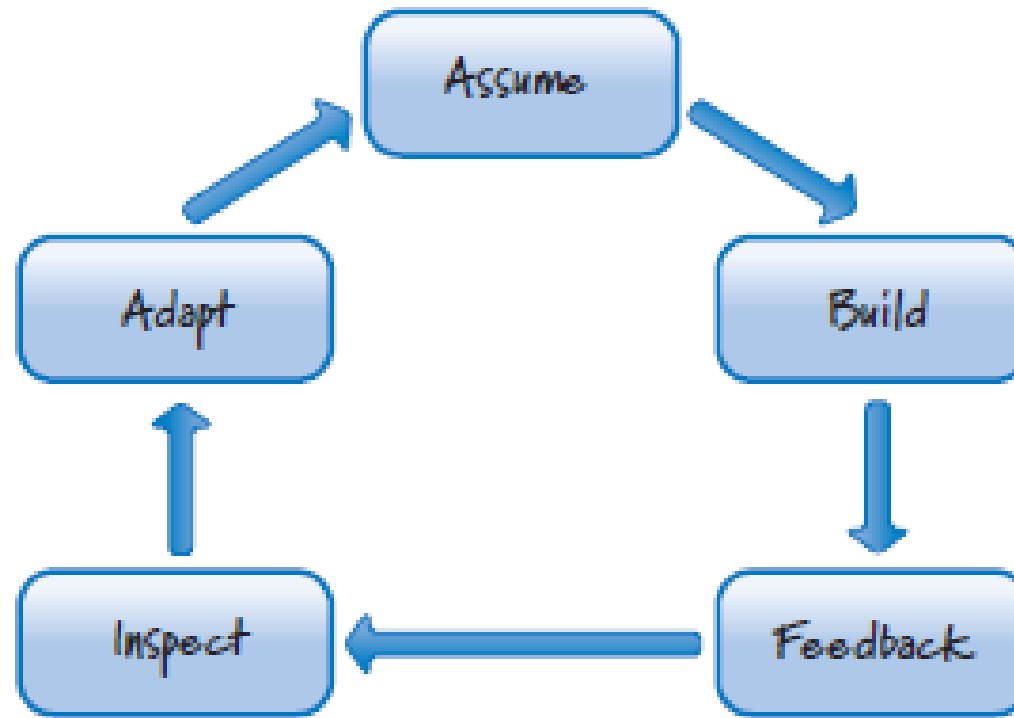
Balance Predictive Up-Front Work with Adaptive Just-in-Time Work



Validated Learning

- Validate important assumptions fast
 - May impact sprint planning
 - May be work with others (marketing, legal, etc.)
- Leverage multiple concurrent learning loops
 - Many loops in scrum – use these as learning opportunities
- Organize workflow for fast feedback

Learning Loop Pattern



Learning loop pattern

Work in Process (WIP)

- WIP = Work that is started but not yet finished.
- Four principles
 - Use economically sensible batch sizes: Aim for smaller, iterative batches rather than doing everything at once
 - Recognize inventory and manage it for good flow: In product development, the inventory is the WIP. Inventories have costs as does WIP
 - Focus on idle work, not idle worker:
 - Idle work is work we want to do but cannot do because something is preventing us
 - Idle workers is the capacity to do more work
 - If we focus on reducing idle workers, we do this by often giving them more work to do, often resulting in idle work.
- Consider cost of delay: Any delay can have an important impact on the project. See example on next slide.

Cost of Delay Example

TABLE 3.3 Example Cost-of-Delay Calculation

Parameter	Value
Duration with full-time documenter	12 months
Duration with documenter assigned at the end (when we reach the state of “all but documented”)	14 months
Cycle-time cost for doing documentation at the end	2 months
Cost of delay, per month	\$250K
Total cost of delay	\$500K
Annual fully burdened cost of documenter	\$90K
Monthly fully burdened cost of documenter	\$7.5K
Cost for full-time documenter	\$90K
Cost for documenter if assigned at end	\$15K
Incremental cost for full-time documenter	\$75K

Progress

- Adapt to real-time information and replan. Remember that the goal is not to stick to the original plan, but adapt to new information that arrives during the development process
- Measure progress by validating working assets.
- Focus on value-centric delivery; not on the artifacts of the development process
- Deliver high-value (from the customer's perspectives) features Sooner

Performance

- Go fast but never hurry
 - Sprints deliver fast, give opportunities for testing fast
 - But as we go through activities specification, design, build we should be precise and not hurry.
- Build-in quality
 - In sequential design, a great deal of responsibility for quality is placed on “testing” – we test in quality. Find flaws or imperfections late – expensive
 - In Scrum quality comes from the team and the iterative nature of development with early and ample testing opportunities
- Employ minimally sufficient ceremony
 - Unlike plan-driven development, Scrum favors continuous development. There is no big product to deliver at the end so there is not big ceremony for celebration

Closing

- For a complete comparison between plan-driven development and agile processes, please see Table 3.4 of the textbook.
- I will go through a summary of this comparison in the class.